

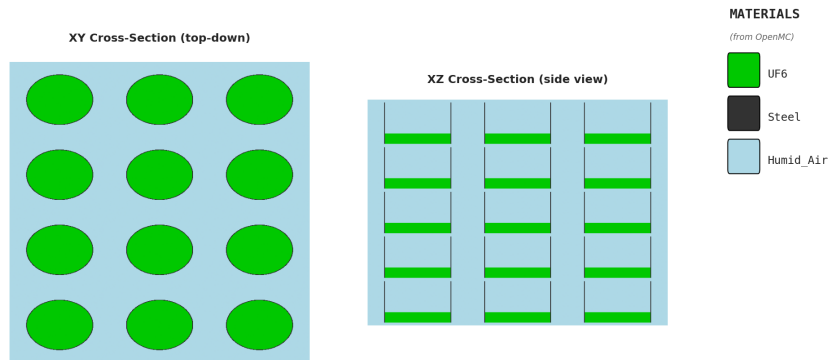
Cylinder Array Independent Review (Assumptions)

- **Report name:** cylinder-array-review
 - **Date:** 2026-02-27
 - **Prepared by:** Crit-Buddy (independent review)
 - **Units:** cm and g/cc
 - **Solver basis:** OpenMC model in `templates/cylinder/openmc/model.py`, reproduced in MCNP
 - **Nuclear data (OpenMC):** ENDF/B-VII.1 HDF5
 - **Temperature:** 293 K
-

1. Overview (What This Case Is)

This is a $3 \times 4 \times 5$ **infinite array** of identical vertical cylinders. Each cylinder has a fissile core (UF6 or UO2F2), a steel wall shell, and flat steel top/bottom caps. The array is surrounded by humid air, with **reflective boundary conditions** on all six outer planes to represent an infinite array.

Three case families are covered: - Dry UF6 fill sweep - Wet UO2F2 fill sweep - UO2F2 H/U sweep at full fill



Geometry plots generated by OpenMC

Figure 1: Geometry Cross-Sections

2. Common Geometry Inputs

Parameter	Value
Rows x Cols x Layers	3 x 4 x 5
Inner radius	12.70
Internal height	100.0
Wall thickness	0.3175
Wall material	Steel (SS316 approximation)
Horizontal gap (wall-to-wall)	12.70
Vertical gap (wall-to-wall)	7.62
Environment	Humid air
Boundary	Reflective
Reflector thickness	6.35

3. Materials Summary

Material	Density	Composition Basis
UF6	5.09	<code>materials.py:create_uf6</code> (U-235/U-238 from wt% + F-19)
UO2F2	<code>uo2f2_density(h_to_u)</code>	<code>materials.py:create_uo2f2</code> (U, O, F, H per H/U)
Steel	8.0	<code>materials.py:create_steel</code> (Fe, Cr, Ni, Mo, Mn)
Humid air	0.0011	<code>materials.py:create_humid_air</code> (N, O, Ar, H)

Thermal scattering: apply `lwtr` (OpenMC `c_H_in_H2O`) for UO2F2 when $H/U > 0$.

UO2F2 density table (H/U sweep)

H/U	Density (g/cc)
0	6.3700
5	3.7804
10	2.8758
15	2.4153
20	2.1364
25	1.9493
30	1.8151
40	1.6354
50	1.5207
75	1.3587
100	1.2736

4. Physics and Run Settings

Parameter	Value
Run mode	Eigenvalue
Particles per batch	10,000
Total batches	150
Inactive batches	50
Active batches	100
Total histories	1,000,000

Source distribution (OpenMC box): - x: -45.0850 to 45.0850 - y: -64.4525 to 64.4525 - z: -241.5100 to 241.5100

MCNP guidance: KCODE 10000 1.0 50 150, with KSRC or SDEF box matching the bounds above.

5. Assumptions and Limits

- Room temperature (293 K).
 - No burnup or depletion.
 - No absorber credit.
 - Flat end caps.
-

6. Source of Truth (Python Models Included)

- `source/cylinder_template.py` (defaults + derived geometry)
- `source/cylinder_openmc_model.py` (geometry + settings)
- `source/materials.py` (materials + densities)